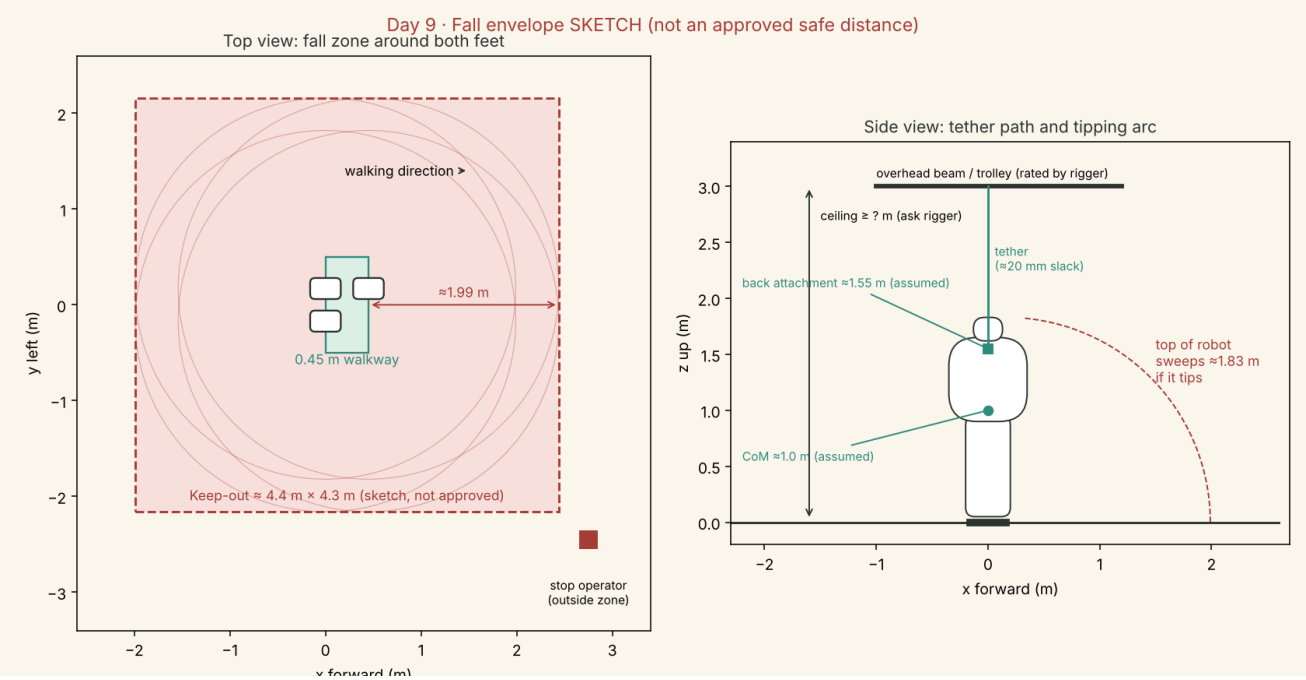
Day 009 — Robot Fall Hazards

Assistive Systems for Trusted Robotic Operations

R0.3 · 2026-11-02 · Owner: Faisal Farouq Day 9 result: PASS (desk work: hazard questions only). No rating, safe distance or anchor is approved here.

The short version

- A 55 kg, 1.83 m robot that tips over can reach about 2 m from its feet and hit the floor with roughly 540 J: the same as dropping 55 kg from 1 m.
- A keep-out zone of about 4.4 × 4.3 m around the 0.45 m walkway is the starting sketch. A rigger and test reviewer set the real one.
- Two different things stop a fall. Balance control (software and motors) keeps the robot upright. Fall arrest (overhead tether) catches it when balance fails, including when power is cut. Each is tested separately.
- The tether has to move with the robot. A fixed overhead point tilts the tether as the robot walks, and a fall at the end of the walkway would swing it back like a pendulum.
- Slack decides the catch load. In an illustrative screen (not for choosing equipment), 20 mm of slack gives ≈1,600 N and 200 mm gives ≈11,300 N.



Inputs

- risk_records/hazards.csv: ASTRO-H-01 (fall or arrest failure), ASTRO-H-14 (tether snag or assistance), ASTRO-H-17 (power removal drops the load), ASTRO-H-20 (friendly look draws people close).
- cad_specifications/parameters.json: 1.829 m, 55 kg, feet 0.32 × 0.22 m, 0.34 m apart.
- requirements/day_006_D03_D04_configuration.md: tether slack 20 mm and CoM 1.0 m (both assumed).
- simulation plan: walkway 0.45 m (three steps of ≈0.15 m).

Numbers

Item	Working	Result	Status
Static weight	55 × 9.81	539.55 N	Target mass
Tipping reach from foot centre	1.829 height + 0.16 half-foot	≈1.99 m	Sketch
Keep-out rectangle	length 0.45 + 2 × 1.99; width 0.34 + 2 × 1.99	≈4.4 m × 4.3 m	Sketch, not approved
Untethered fall energy	55 × 9.81 × 1.0 m CoM drop	≈540 J	CoM assumed
Catch force, illustrative only	m·g·(1 + slack ÷ stretch), stretch 0.01 m	20 mm → 1,619 N · 50 mm → 3,237 N · 200 mm → 11,331 N	Not for equipment selection (ignores rope stiffness, peaks, geometry)

The last row shows why slack matters, nothing more. The rigger calculates the real peak load.

Tether angle if the overhead point does not move

Back attachment ≈1.55 m above the floor (assumed). Tether length L = ceiling anchor height − 1.55 m.

Anchor height	L	Anchor at walkway start (robot moves 0.45 m)	Anchor over the walkway middle (±0.225 m)
2.6 m	1.05 m	23.2°	12.1°
3.0 m	1.45 m	17.2°	8.8°
3.5 m	1.95 m	13.0°	6.6°

Recommendation to put to the rigger: a rated trolley on a beam or track along the walkway, so the tether stays vertical. If a fixed point is the only option, put it over the walkway middle with the highest ceiling available, and record the angle.

The arrest path: one question per part

Part	Question	Evidence to get	Who answers
Robot attachment point	Where on the frame (not the shell) does the tether connect, and what load can that bracket take?	Bracket drawing with rated load	Mechanical reviewer (Day 199)
Tether / lanyard	What length, material, slack and stretch? What drop before it catches?	Datasheet; slack set at install	Rigger
Connectors	Which shackles or carabiners, and their working load limit?	Marked ratings; inspection record	Rigger
Trolley or overhead point	Fixed point or trolley along the walkway? Rated load?	Product rating; install record	Rigger
Beam / frame	Is the beam or portable gantry rated for the arrest load?	Rating plate or engineer's letter	Rigger / building owner
Building anchor	What is it fixed into, and who approved it?	Written approval from the facility	Facility owner
After a catch	How do you lower and recover a 55 kg robot hanging on the tether?	Written recovery steps; second person	Rigger + you

Note: fall gear made for people (harnesses and lanyards, ANSI/ASSP Z359 family) is designed and tested for people. It is not assumed suitable for a robot. Use it only if the rigger approves that exact use in writing.

Site checklist (fill in on Day 134)

Item	Need (starting value)	Measured
Clear floor area	≥ 4.4 m × 4.3 m plus a walkway for people outside it	
Ceiling or beam height	Ask rigger; the table above assumes 2.6–3.5 m	
Overhead structure	Beam, track or gantry the rigger can rate	
Floor	Flat, hard, dry; note the material (friction test, Day 117)	
Power outlets	Within cable reach, outside the keep-out zone	
Barrier	Rope or panels at the keep-out line	
Spectator line	Behind the barrier; remote screen if people want to watch (ASTRO-H-20)	
Stop operator position	Outside the zone, with the e-stop in reach and a clear view of the robot	
E-stop reach	Wired e-stop cable ≥ 2.5 m, or a rated wireless stop	
Lighting and cameras	Room for 2 fixed cameras (side, front) outside the zone	

How each hazard is covered

Hazard	What today adds	Still needed
ASTRO-H-01 fall / arrest failure	Keep-out sketch; arrest-path questions	Rated arrest design and inspection (Days 135–137, 258)
ASTRO-H-14 tether snag / assistance	Trolley vs fixed-point angles; load cell in the tether to show how much it helps (Day 146)	Rigger's routing; acceptable assistance set before gait tests (Day 281)
ASTRO-H-17 power loss drops the robot	Arrest must hold with all power off; the robot's motors are never trusted to hold it	Power-off test on the support (Day 235)
ASTRO-H-20 people walk up to it	Barrier, spectator line, stop operator outside the zone	Briefing script before each test

Outside help

When	Who	What to get
Day 135	A qualified rigger or a structural engineer	Written arrest design: anchor, trolley, tether, slack, ratings, inspection
Day 134	Owner or manager of a test space	Written permission, hours, ceiling or beam details, cost

No one is named yet (D-02).

Checks

Check	Result
Fall envelope drawn and labelled as a sketch	PASS
Arrest and balance control described separately	PASS
Every arrest part has a question, the evidence needed and who answers	PASS
No anchor capacity, safe distance or arrest rating presented as approved	PASS

Sources: R0.1 hazard register (ASTRO-H-01, 14, 17, 20); R0.1 engineering baseline p. 8 (the energy screen m·g·(1 + h/d), which R0.1 notes must not be used to select equipment); Day 6 R0.3 assumptions A6-01 and A6-12.